Module 10 Lab – Unsupervised Learning

Objective: Apply K-Means clustering to identify groups in synthetic customer data and Principal Component Analysis (PCA) to reduce the Iris dataset from four dimensions to two.

Part 1: Supervised vs. Unsupervised Learning

Supervised learning uses features (X) and known target labels (y) to learn predictions. Unsupervised learning uses features (X) without known target labels to discover patterns. Clustering groups similar observations, while dimensionality reduction summarizes the data using fewer dimensions.

Part 2: K-Means Clustering

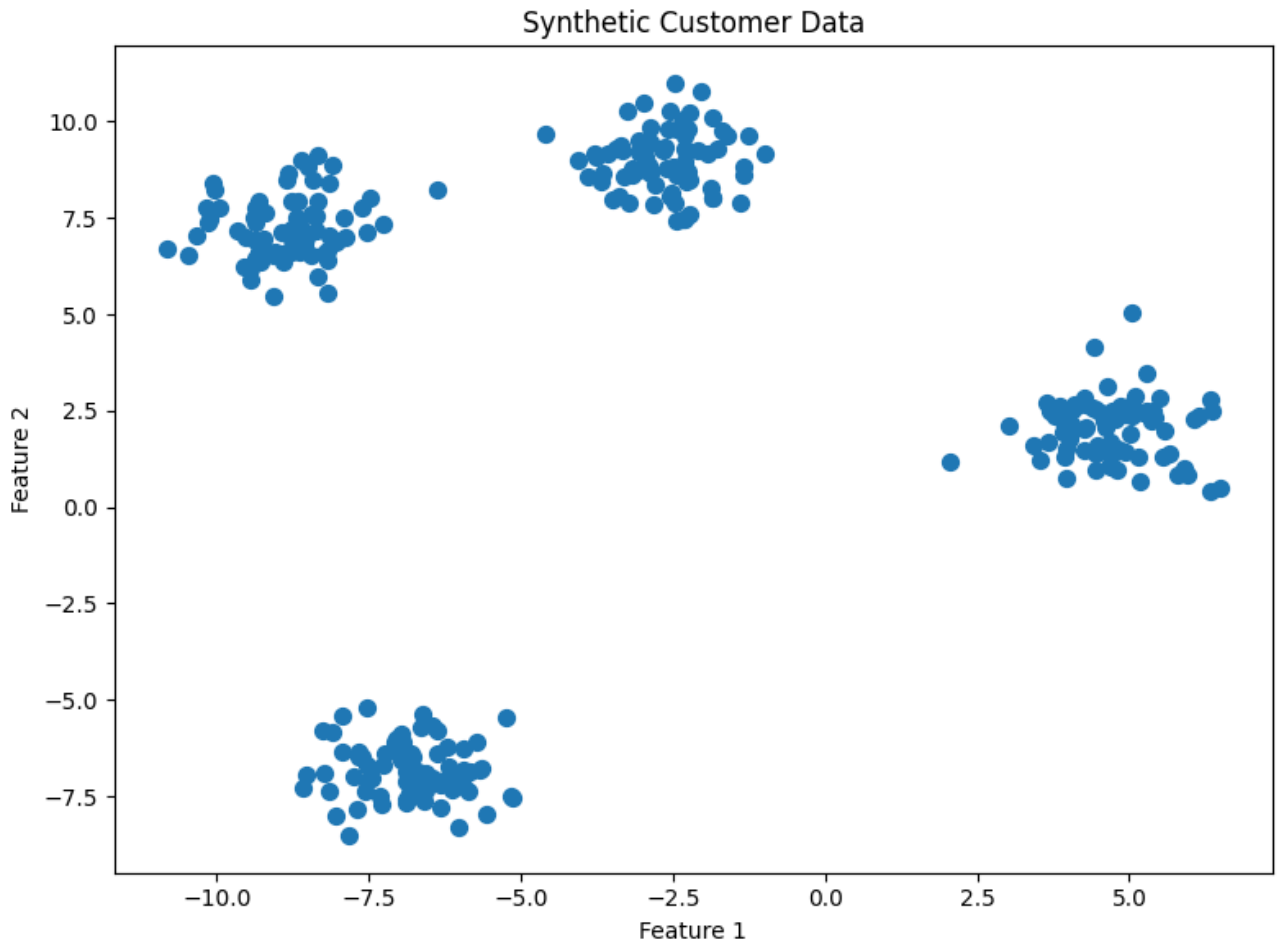
K-Means selects k cluster centers, assigns each observation to its closest center, updates the centers using the assigned observations, and repeats until convergence. Here we use a synthetic dataset representing customer-spending patterns.

Code cell 3

```
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import seaborn as sns
from sklearn.datasets import make_blobs, load_iris
from sklearn.cluster import KMeans
from sklearn.decomposition import PCA

# Create a repeatable synthetic dataset with four underlying groups.
X, y = make_blobs(n_samples=300, centers=4, cluster_std=0.8, random_state=42)

plt.figure(figsize=(8, 6))
plt.scatter(X[:, 0], X[:, 1], s=50)
plt.title("Synthetic Customer Data")
plt.xlabel("Feature 1")
plt.ylabel("Feature 2")
plt.tight_layout()
plt.show()
print(f"Generated {X.shape[0]} observations with {X.shape[1]} features.")
```




```
Generated 300 observations with 2 features.
```

Task 1: Find the Optimal Number of Clusters (Elbow Method)

Fit K-Means for k=1 through k=10 and plot each model's inertia, which measures within-cluster squared distances. Look for a bend where adding clusters yields smaller improvements.

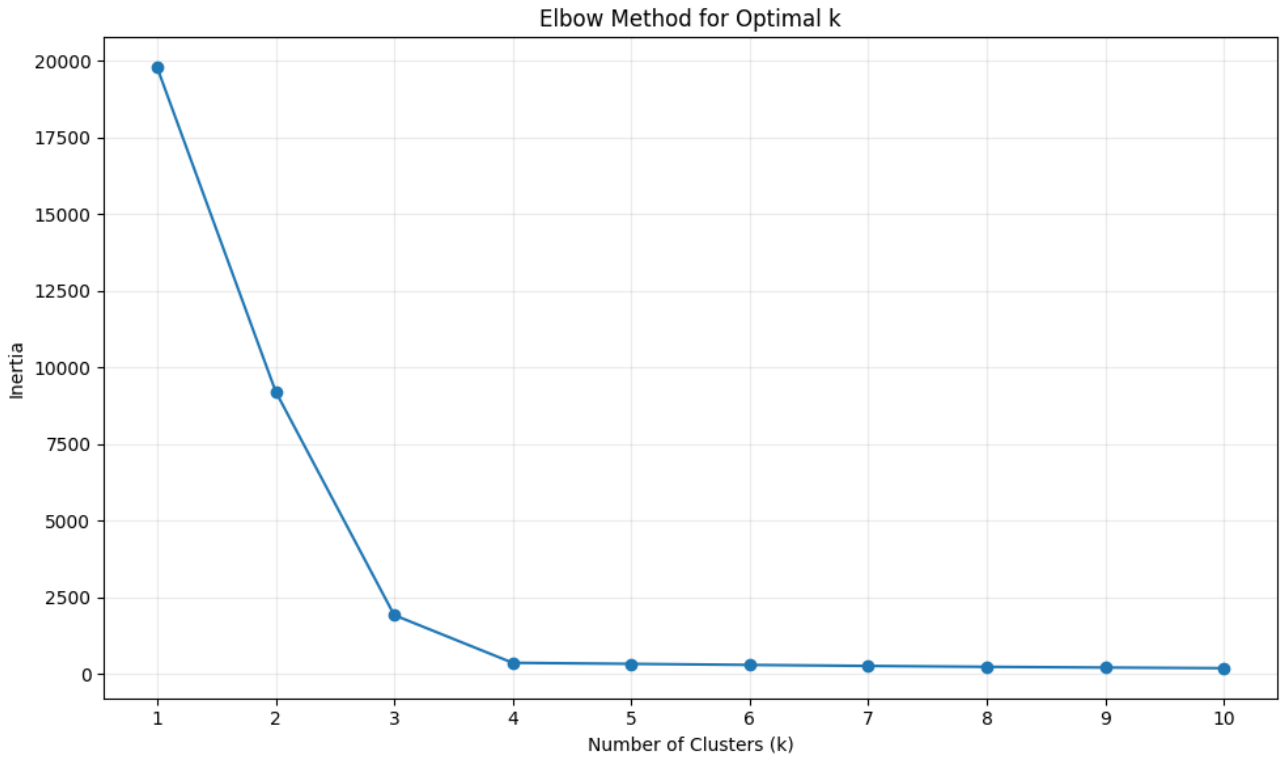
Code cell 5

```
  inertia = []
  for k in range(1, 11):
      kmeans = KMeans(n_clusters=k, random_state=42, n_init=10)
      kmeans.fit(X)
      inertia.append(kmeans.inertia_)

  print(pd.DataFrame({"k": range(1, 11), "Inertia": np.round(inertia,
      2)}).to_string(index=False))
  plt.figure(figsize=(10, 6))
  plt.plot(range(1, 11), inertia, marker="o")
  plt.xticks(range(1, 11))
  plt.title("Elbow Method for Optimal k")
  plt.xlabel("Number of Clusters (k)")
  plt.ylabel("Inertia")
  plt.grid(alpha=0.25)
  plt.tight_layout()
  plt.show()
  print("Interpretation: The curve bends near k=4, so four clusters are a reasonable choice.")
```

```
k  Inertia
 1 19780.25
 2  9211.21
 3  1919.36
```

```
 4   362.47
 5   329.27
 6   294.61
 7   261.56
 8   232.01
 9   209.10
10   188.72
```




```
Interpretation: The curve bends near k=4, so four clusters are a reasonable choice.
```

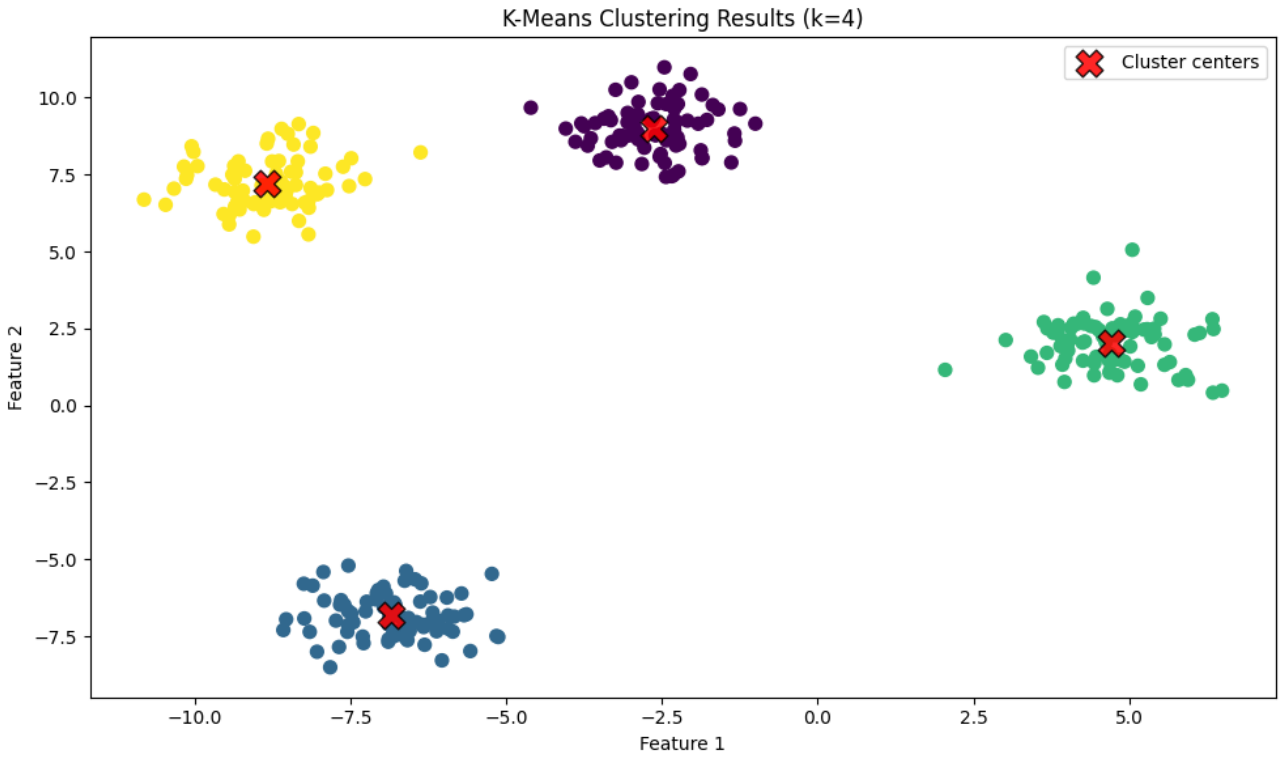
Task 2: Perform K-Means Clustering and Visualize

Fit k=4 and plot each point by its predicted cluster assignment. Red X markers identify the computed centers.

Code cell 7

```
kmeans = KMeans(n_clusters=4, random_state=42, n_init=10)
kmeans.fit(X)
labels = kmeans.labels_
centers = kmeans.cluster_centers_

plt.figure(figsize=(10, 6))
plt.scatter(X[:, 0], X[:, 1], c=labels, s=50, cmap="viridis")
plt.scatter(centers[:, 0], centers[:, 1], c="red", s=220, alpha=0.85,
            marker="X", edgecolors="black", label="Cluster centers")
plt.title("K-Means Clustering Results (k=4)")
plt.xlabel("Feature 1")
plt.ylabel("Feature 2")
plt.legend()
plt.tight_layout()
plt.show()
print("Number of points in each cluster:")
print(pd.Series(labels).value_counts().sort_index().rename_axis("Cluster").to_string())
```




```
Number of points in each cluster:
Cluster
0    75
1    75
2    75
3    75
```

Part 3: Principal Component Analysis (PCA)

PCA finds new, uncorrelated directions that capture decreasing amounts of data variance. We reduce the Iris dataset from four measured features to two components so the patterns can be visualized on a 2D scatter plot.

Task 3: Apply PCA to the Iris Dataset

Fit two PCA components, report how much variance they retain, and plot the projected observations colored by Iris species. Species labels are used only to color the plot, not to fit PCA.

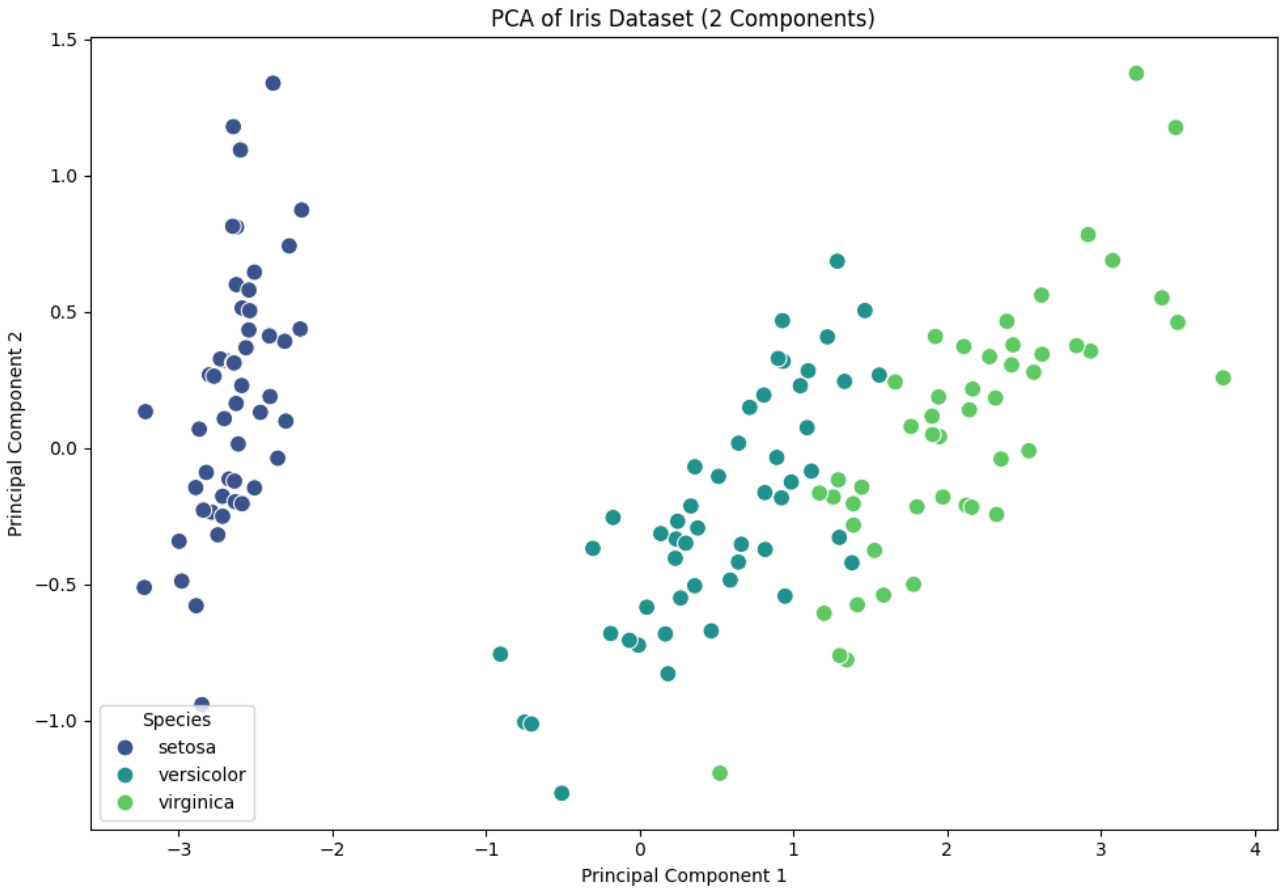
Code cell 10

```
iris = load_iris()
X_iris = iris.data
y_iris = iris.target

pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_iris)
print(f"Explained variance by component: {pca.explained_variance_ratio_}")
print(f"Total variance explained by 2 components: {pca.explained_variance_ratio_.sum():.2%}")

pca_df = pd.DataFrame(X_pca, columns=["Principal Component 1", "Principal Component 2"])
pca_df["Species"] = [iris.target_names[i] for i in y_iris]
plt.figure(figsize=(10, 7))
sns.scatterplot(data=pca_df, x="Principal Component 1", y="Principal Component 2",
                hue="Species", palette="viridis", s=85)
plt.title("PCA of Iris Dataset (2 Components)")
plt.tight_layout()
plt.show()
print(f"Data shape before PCA: {X_iris.shape}; after PCA: {X_pca.shape}")
```

```
Explained variance by component: [0.92461872 0.05306648]
Total variance explained by 2 components: 97.77%
```




```
Data shape before PCA: (150, 4); after PCA: (150, 2)
```

Knowledge Check

1. What is the fundamental difference between supervised and unsupervised learning?

Supervised learning trains using labeled examples, so it knows the correct target answer for each training observation. Unsupervised learning does not use target labels during training; it looks for patterns such as groups or important directions in the features.

2. In the K-Means elbow plot, why not choose the largest possible k to get the lowest inertia?

Inertia naturally decreases as more clusters are added, even if those extra clusters are not meaningful. A very large k can split natural groups into unnecessarily small groups, making the result harder to understand. The elbow balances lower inertia with a simpler, more useful number of clusters; here it occurs around four.

3. What does it mean that the first two PCA components explain over 95% of the variance?

It means two combinations of the original four measurements preserve more than 95% of the dataset's total variance. Therefore a 2D plot keeps most of the original variability and can reveal much of the dataset's structure, although some information is lost. High explained variance does not, by itself, guarantee perfect species separation.